

Engine documentation v3

Došlap engine v3 „Kapacitní“ — load management against each runner’s own demonstrated capacity, and the plan for the Trénink (training guidance) tab. Plan, design, formulas, calibration and evidence.

Status	Phases 1 and 2 implemented and tested (25 Sep 2026). Phase 3 planned.
Engine	v3 = the calibrated sensitive mechanics of v2 + a new, capacity-based load axis (Zátěž). Stored version suffix “-c” (e.g. v0.7.0-c).
Switch	Data a propojení → Engine hodnocení → Kapacitní. v1 (Standardní) and v2 (Citlivý) stay available. Switching to v3 adds the Trénink tab next to Dnes.
Positioning	Training-load monitoring and decision support, not a medical device (MDR). Recommendations are guardrails; pain is routed to a physiotherapist, never to a diagnosis.

1. Why a capacity model

The v1/v2 load axis is a set of about fifteen independent signals (single-session spike, latent spike, pace spike, ACWR, high-intensity ratio, load creep, monotony, descent spike, steep-descent spike, decoupling, HRV, resting HR, HRV variability, fitness–fatigue balance, load × recovery, taper). Several of them fire on the same event — one long hilly run can trigger the session spike, the descent spike and the steep-descent spike — while none of them answers the question a runner actually has: *how much can I do today?*

Engine v3 replaces them with one per-runner **capacity model** used in two views: the Zátěž axis scores how far recent training went beyond what the runner has *demonstrated they tolerate*, and the Trénink tab (Phase 2) spends the remaining headroom. Both views read the same numbers, so they cannot contradict each other: if the guidance says “max 14 km” and the runner does 18, the Zátěž axis shows exactly that overshoot. This follows the load–capacity framework of Bertelsen et al. (2017): injury occurs when cumulative load exceeds the capacity of a structure, and capacity itself varies with recovery.

Decisions taken with the product owner (25 Sep 2026)

- Load management becomes a primary focus, with stronger weight on **jumps in volume and intensity** and on **elevation, especially descent**.
- Capacity is computed from the runner’s own past runs (demonstrated tolerance), personalised by recovery and injury history.
- The quadrant stays the first thing on the Dnes tab; capacity appears lower on Dnes and in full on Zátěž.
- Guardrails now; an AI coach later.
- A separate engine (v3) switched on the Data page, which also enables the Trénink tab.
- Intensity in **HR zones**, plus pace together with HR as **relative effort** against the runner’s last ~2 months (below / at / above usual), similar in spirit to Strava’s Relative Effort.
- Pain override in guidance only when pain is **above 5/10 and** the engine’s own score already calls for a physio referral; lower pain modifies the session instead of cancelling it.

2. Phases

Phase	Scope	Status
1 — Capacity engine	Engine v3 switch; capacity model (5 channels, per run and per 7 days, decay, tolerance filter); readiness scaling; capacity-based Zátěž axis with co-occurrence scoring and latent memory; load weight 0.30 → 0.40; HR zones and Z4+ minutes (HR histogram stored with detailed streams); relative effort and HR-at-pace vs 8 weeks; Zátěž capacity panel; compact capacity strip on Dnes (below the quadrant); documentation, tests, calibration on demo runners.	Done
2 — Trénink tab (guardrails)	Daily session type (default inferred from history, switchable); km target + ceiling; HR-zone and pace range; Z4+ cap; ascent/descent ceilings; terrain; reasons. Weekly budget and progression (deload, taper), 48-hour rule for hard sessions, readiness after the morning sync (provisional state), overrides (pain > 5 + physio referral, pain 3–5 modifies, mechanics drift, active injury). v3 version of the sensitivity sandbox.	Done
3 — Personalisation & validation	Personal recovery cost (next-morning HRV / resting HR response per load unit); refined tolerance learning; log of guidance vs. what was run and the outcome (OSTRC-H2 labels); calibration on real data; groundwork for the AI coach.	Planned

3. The capacity model

3.1 Channels

Each channel is scored per single session and per rolling 7 days. Evidence grades use the app's existing scale (A consistent prospective evidence in runners; B mixed or emerging; C mechanistic).

Channel (UI)	Exposure per session	Weight	Grade	Floor run / week	Main evidence
Objem	running km	1.0	B	3 km / 10 km	Frandsen 2025; Nielsen 2014
Intenzita	minutes at ≥ 80 % of heart-rate reserve (Z4+)	0.9	B	5 / 10 min	Neal 2024; Paquette 2020
Klesání	descent m × (1 + 3.2 × grade)	0.8	C	50 / 150 m	Gottschall & Kram 2005; McHugh 2003; Vernillo 2017
Stoupání	ascent m	0.5	C	50 / 150 m	Minetti 2002; Vernillo 2017
Celková zátěž	Banister TRIMP, all sports	0.7	B	30 / 100 AU	Banister 1991; Paquette 2020

The descent weighting uses the rise of impact force with downhill grade reported by Gottschall & Kram (2005) — normal impact peaks +32 % at −6° and +54 % at −9°, i.e. roughly +3.2 % per 1 % of grade. Steep descending therefore counts more per metre of drop. Without an elevation profile the runner's own median steepness factor is used (default 1.16 ≈ a −5 % descent).

3.2 Capacity = demonstrated tolerance

per-run capacity = max over tolerated sessions in the last 90 days of exposure × decay(age)
 decay = 1 for age ≤ 30 d, then $0.5^{((\text{age} - 30) / 30)}$ (30-day half-life)
 needs ≥ 3 sessions carrying the channel in the last 30 days, else "unknown"
 weekly capacity = max(average week of days 7–34 ago,
 0.9 × best pain-free rolling 7-day window ending 7–41 days ago, floor)
 needs ≥ 4 weeks of history
 tolerated session = no running-relevant pain ≥ 3/10 (check-in, run rating) and no niggle
 on the day of the run or the 3 days after

The single-run reference is exactly the one used by the largest running cohort to date: Frandsen et al. (2025) followed 5,205 runners (588,071 sessions) and found that a run more than 10 % longer than the longest run of the previous 30 days raised the overuse-injury rate (HRR 1.64 for +10–30 %, 1.52 for +30–100 %, 2.28 for > +100 %). The tolerance filter is what only Došlap can add: a run followed by pain was, by definition, not tolerated and does not raise capacity.

3.3 Readiness and injury history

parts (0–1) for the night / day: $HRV = \text{clamp}(-z / 2)$ z vs the runner's days 8–35 before
RHR = $\text{clamp}(+z / 2)$
sleep = $\text{clamp}((\text{usual} - \text{that night}) / 2 \text{ h})$
soreness $\geq 6 \rightarrow (s - 5)/5$; fatigue $\geq 6 \rightarrow (f - 5)/5$ (check-in)
readiness = $1 - 0.3 \times \text{mean}(\text{parts}) \rightarrow 0.70 \dots 1.00$
margins: per run 10 %, per week 15 %, $\times (1 - 2.5 \times (\text{frailty} - 1))$
frailty 1.04–1.20 for an injury in the last 12 months $\rightarrow$ margins down to 50 %

Readiness scales capacity down: a normal run after a night with suppressed HRV, raised resting HR and short sleep counts as an exceedance, the same run after a good night does not. HRV-guided monitoring of this kind follows Plews et al. (2013). In v3 injury history acts through the margins only; the v1/v2 multiplier on the load score is not applied, so it is not counted twice.

4. The v3 Zátěž axis

ratio_run = exposure / (per-run capacity before that day $\times$ readiness that day) worst run of 7 days
ratio_week = 7-day sum / (weekly capacity $\times$ mean readiness of the 7 days)
points(r, m) = 0 $r \leq 1 + m$
= linear to 6 at 1.3 $\times$ (mild)
= linear to 20 at 2.0 $\times$ (moderate)
= linear to 40 at 3.0 $\times$, capped (high — RUNSAFE's clear danger zone)
latent = runs 7–27 days old with $r > 1.3$: $\min((r - 1.3) \times (28 - \text{age}) / 21 \times 22, 16)$
channel = weight $\times \max(\text{points_run}, \text{points_week}, \text{latent})$
Zátěž = Σ channels sorted high $\rightarrow$ low $\times (1, 0.5, 0.25, 0.25, 0.25)$
+ pace spike (≤ 10) + monotony (≤ 12) + race-proximity taper flag
overall = $0.38 \times \text{mechanics} + 0.40 \times \text{Zátěž} + 0.52 \times \text{symptoms}$ (v1/v2: 0.30 for Zátěž)

Co-occurrence instead of stacking. The worst channel counts fully and each additional channel that also exceeds capacity adds 50 % / 25 %, so a run that is long *and* hilly *and* hard scores above any one of them alone, without one event being counted three times. The latent term keeps a big jump visible for 1–4 weeks, when the IOC consensus (Soligard et al., 2016) places the peak of injury likelihood after a rapid load rise.

Stronger weight on real jumps. A single run at 1.5 $\times$ capacity scores 10 points in v3 (4 in v1); at 2.5 $\times$ it scores 30 and flips the quadrant on its own. At the same time the noise sources are removed: ACWR stays context only (Impellizzeri et al., 2020, and no predictive value in runners in Frandsen 2025), HRV, resting HR and sleep move into readiness, and the descent / steep-descent spikes, load creep, fitness–fatigue balance and HRV variability are replaced by the channels.

5. Intensity: HR zones and relative effort

Zone	Z1	Z2	Z3	Z4	Z5
% of heart-rate reserve (Karvonen)	50–60	60–70	70–80	80–90	90–100

HR max = $\max(208 - 0.7 \times \text{age}$ (Tanaka et al., 2001), hardest observed average HR + 10); resting HR = median of nightly values; zones from the heart-rate reserve (Karvonen et al., 1957). Minutes in Z4+ come from the HR histogram (2-bpm bins) that is now stored with every detailed Garmin stream; without a stream they are estimated from the run's thirds or average HR assuming a normal spread (4 % / 6 % of HRR), so a tempo run that averages just under Z4 still counts its hard minutes. The intensity-distribution context is the ~80/20 low/high pattern of endurance athletes (Seiler, 2010).

Relative effort (pace together with HR)

- **Per run:** TRIMP compared with the runner's runs of the previous 56 days — below the 25th percentile “pod obvyklým”, 25th–75th “obvyklé”, above “nad obvyklým”, above the 8-week maximum “výrazně nad” (≥ 6 prior runs).

- **HR at pace:** a Theil–Sen line of HR against speed on the runner's flat-ish runs 7–56 days earlier (≥ 8 runs, ≥ 0.15 m/s spread) gives the expected HR for this run's pace; $|\Delta\text{HR}| \geq 5$ bpm is shown. A run needing +8 bpm for the usual pace hints at fatigue, heat or illness; a negative value at equal pace indicates improving fitness.
- **Per week:** the last 7 days' TRIMP against the 25th–75th percentile of rolling weeks over the previous 8 weeks.
- Strava's Relative Effort is a zone-weighted TRIMP with a weekly range from past weeks (Strava Help Center). Došlap uses the runner's own 8-week distribution and adds HR at pace, which Strava does not show.

6. Calibration and verification (Phase 1)

All twelve seeded demo runners were replayed day by day over 90 days through v1 and v3 (same data, engine only changed). P = share of days with Zátěž ≥ 25 ("Přetížení" or "Kritická kombinace").

Runner (demo)	P v1	P v3	mean v1	mean v3	main v3 driver
Adéla Kučerová	0.13	0.03	12.5	8.1	descent
Marek Beneš	0.30	0.30	16.0	14.0	descent
Lucie Horáková	0.44	0.14	18.9	10.6	intensity
Tomáš Řehák	0.31	0.09	18.6	10.3	descent
Veronika Šimková	0.02	0.00	1.8	1.7	—
Pavel Urban	0.18	0.08	14.6	9.9	intensity
Klára Bártová	0.20	0.12	13.4	9.3	descent / intensity
Jiří Málek	0.39	0.23	23.6	18.4	descent
Nikola Pešková	0.22	0.17	13.9	12.2	descent
Radek Souček	0.61	0.01	28.5	13.9	monotony
Tichý drift	0.14	0.00	9.9	2.7	— (mechanics drift only, as designed)
Kritické přetížení	0.30	0.19	20.1	14.6	volume, intensity, systemic

- v3 removes most "Přetížení" days that v1 produced from noise (fitness–fatigue balance, monotony, steep-descent spikes), e.g. Radek 61 % $\rightarrow$ 1 %, Lucie 44 % $\rightarrow$ 14 %; days with genuine exceedances remain.
- Descent is now the leading driver for the hilly runners, and intensity jumps are caught earlier: Klára's two harder weeks (69 min in Z4+ against a weekly capacity of 34) show as "Přetížení" in v3, which v1 missed.
- Both demo accounts keep their intended state today: Tichý drift = silent drift, Kritické přetížení = critical (v3 Zátěž 66: weekly volume 3.4 $\times$ capacity, intensity 3.1 $\times$, readiness 77 %).
- Tests: 157 passing after Phase 2 (Phase 1 added 9 capacity tests; Phase 2 added 12 guidance tests — overrides, pain modification, ceilings, 48-hour rule, taper, deload, provisional state, paces, exhausted budget — and a v3 sandbox parity test). Live assessment ≈ 0.16 s; a 6-month daily history replay costs ≈ 10 % more than v2. The Zátěž, Dnes, Trénink and sandbox views were checked in a browser at desktop and phone width.
- *Limitation:* the demo runners are synthetic. Weights, margins and bands are expert-set and must be re-calibrated on real runners and real injury reports in Phase 3.

7. Phase 2 — Trénink tab (implemented)

A guardrail, not a coach: it gives safe ranges for the day; it does not write a periodised plan (an AI coach may follow later). The tab appears next to Dnes when the runner is on engine v3. Guidance is computed on every live recompute (Garmin sync, check-in, run rating, a new day) and stored with the assessment; history replays skip it. Code: backend/app/metrics/guidance.py, frontend/src/training.tsx.

Element	Rule
Session type	Default inferred from history (usual run days, long-run weekday, hard days); the runner can switch between volno / regenerace / lehký / dlouhý / kvalitní; each type shows its limits.
Distance	Target range and hard ceiling; long run $\leq$ per-run ceiling and $\leq \sim 30$ % of the weekly budget.

Element	Rule
Intensity	HR-zone range and the matching pace range from the runner's own HR-pace relation; cap on minutes in Z4+; a quality session only if ≥ 48 h since the last hard one, readiness ≥ 0.85 , intensity budget left, no pain ≥ 3 , Zátěž < 25 and no mechanics drift.
Elevation / terrain	Ascent and descent ceilings from the channel capacities; flat or soft surface when descent headroom is gone or mechanics drift.
Weekly budget	Weekly capacity $\times$ progression (1.10 fresh and low load, 1.05 default, 1.00 when load rises or readiness < 90 %) $\times$ today's readiness; $\times 0.7$ deload after three full weeks or when Zátěž ≥ 25 ; taper towards the goal race; left today = budget – last 6 days – today (Seiler 2010 for the easy/hard balance).
Timing	Recomputed on every sync, check-in and run rating; marked provisional until today's sleep and HRV arrive (the pre-07:00 Garmin auto-sync makes it ready at wake-up).
Overrides (as decided)	Pain $> 5/10$ and the engine's own physio referral (critical quadrant, alert tier or silent drift $\rightarrow$ triage physio_48h / physio_7d) $\rightarrow$ no run, book the physio. Pain 3–5 $\rightarrow$ modify (shorter, easier, flat/soft, no quality). Pain > 5 without a referral $\rightarrow$ regeneration or cross-training, no physio override. Mechanics drift $\rightarrow$ cap intensity and descent. Active injury (OSTRC) $\rightarrow$ rest.
Explanation	2–4 reasons tied to the signals (e.g. sleep -1.2 h, HRV below normal, last hard session yesterday, 82 % of this week's volume budget used).

Implementation details

Rule	As implemented
HR bands per type	regenerate 50–65 % HRR (Z1–Z2), lehký 60–72 % (Z2), dlouhý 60–75 % (Z2), kvalitní 80–92 % in the hard segments (Z4–Z5), warm-up and cool-down Z1–Z2.
Pace	From the runner's own HR-speed line (Theil–Sen, flat runs of the last 8 weeks) — used only when the resulting paces fall inside the speeds the runner actually runs (± 15 %); otherwise from their usual easy pace (lehký 0.97–1.05 $\times$, regenerate 1.06–1.15 $\times$, dlouhý 1.00–1.08 $\times$). Quality sessions are paced by HR.
Limits	Every type's km $\leq$ min(per-run volume ceiling, what is left of the weekly budget). The per-run ceiling is the capacity threshold of the Zátěž axis, so running exactly the ceiling adds no load points (verified by a test). Z4+ in a quality session $\leq$ min(intensity ceiling, 45 min), halved under mechanics drift; descent limits halved with pain 3–5 or drift.
Default type	Taken from the last 8 weeks: usual run days (≥ 40 % of weeks), long-run weekday (the week's longest run $\geq 1.25\times$ the others, same weekday in $\geq$ half of such weeks), hard days (runs with ≥ 10 min in Z4+). Readiness < 80 % $\rightarrow$ regeneration run; < 2 km left in the weekly budget $\rightarrow$ rest; long run only if its target $\geq 1.2\times$ the usual easy run.
Taper / deload	8–14 days before the goal race volume $\times 0.7$, ≤ 7 days $\times 0.5$ and no long run, following the 2-week, 41–60 % volume-reduction taper of Bosquet et al. (2007). Deload $\times 0.7$ when Zátěž ≥ 25 or after three consecutive 7-day windows each ≥ 90 % of weekly capacity.
Already run today	The tab shows what was done and the limits still left for the day.

Guidance for the demo runners (25 Sep 2026)

Runner (demo)	Recommended today	Main reason
Kritické přetížení	Volno + book the physio	pain 8/10 and the engine's 48-hour physio referral
Jiří Málek	Volno + book the physio	pain 7/10 and a physio referral; readiness 71 %
Adéla Kučerová	Lehký běh 4.9–6.4 km, Z2	pain 4/10 modifies the day (no long run or quality), mechanics drift $\rightarrow$ flat or soft surface
Tichý drift	Dlouhý běh 8.8–10.4 km, Z2	usual long-run day (Friday), 54 % of the weekly budget used
Marek Beneš	Kvalitní trénink, 27–45 min in Z4+	5 days since the last hard session; deload week $\times 0.7$
Pavel Urban	Lehký běh 6.3–8.2 km, Z2	fresh and low load $\rightarrow$ build week $\times 1.10$
Klára Bártová	Volno	weekly volume budget used up (98 %), pain 5/10
Tomáš Řehák, Radek Souček	Volno	7-day volume at 214 % / 145 % of the weekly budget

Sensitivity sandbox (v3)

The Citlivostní analýza sandbox now models either engine. In the Kapacitní mode its load knobs are each channel's per-run and 7-day ratio to capacity plus a readiness knob that rescales them; the v1/v2-only load knobs are hidden. Seeded with "Načíst moje data", it reproduces a v3 runner's live Zátěž and Mechanika scores exactly (checked by a test and in the browser: 66 and 29 for the Kritické přetížení demo).

8. Phase 3 — personalisation and validation

- **Personal recovery cost:** how much next-morning HRV and resting HR move per unit of load in each channel (needs 8–12 weeks of nightly HRV; shrunk towards a default slope while data are thin).
- **Tolerance learning:** refine which sessions count as tolerated using run ratings, niggles and OSTRC-H2 reports (Clarsen et al., 2020).
- **Guidance vs. actual log** and outcomes, so it can be shown whether respecting the ceilings reduces spikes and pain; re-calibration of weights, margins and bands on real data.
- Groundwork for the AI coach (later).

9. Limitations and positioning

- Došlap is not a medical device. The engine supports decisions of runners and physiotherapists; it does not diagnose or prescribe treatment.
- The core evidence is observational (prospective cohorts). A ceiling is a guardrail, not a guarantee; training below capacity also has costs (Gabbett, 2016).
- The intensity and systemic channels need heart rate; descent steepness needs an elevation profile. Estimation falls back gracefully but is less precise.
- Risk factors for running injury are multifactorial (Hulme et al., 2017); load is one modifiable part of them.

References

All references were checked against the publisher, PubMed or institutional repository records on 25 Sep 2026.

- [1] Banister EW. Modeling elite athletic performance. In: MacDougall JD, Wenger HA, Green HJ (eds). *Physiological Testing of the High-Performance Athlete*. 2nd ed. Champaign, IL: Human Kinetics; 1991:403–424.
- [2] Bertelsen ML, Hulme A, Petersen J, Brund RK, Sørensen H, Finch CF, Parner ET, Nielsen RO. A framework for the etiology of running-related injuries. *Scand J Med Sci Sports*. 2017;27(11):1170–1180. doi:10.1111/sms.12883
- [3] Bosquet L, Montpetit J, Arvisais D, Mujika I. Effects of tapering on performance: a meta-analysis. *Med Sci Sports Exerc*. 2007;39(8):1358–1365.
- [4] Clarsen B, Bahr R, Myklebust G, et al. Improved reporting of overuse injuries and health problems in sport: an update of the Oslo Sport Trauma Research Center questionnaires. *Br J Sports Med*. 2020;54(7):390–396.
- [5] Foster C, Florhaug JA, Franklin J, et al. A new approach to monitoring exercise training. *J Strength Cond Res*. 2001;15(1):109–115.
- [6] Frandsen JSB, Hulme A, Parner ET, Møller M, Lindman I, Abrahamson J, Sjørup Simonsen N, Sandell Jacobsen J, Ramskov D, Skejød S, Malisoux L, Bertelsen ML, Nielsen RO. How much running is too much? Identifying high-risk running sessions in a 5200-person cohort study. *Br J Sports Med*. 2025;59(17):1203–1210. doi:10.1136/bjsports-2024-109380
- [7] Gabbett TJ. The training–injury prevention paradox: should athletes be training smarter and harder? *Br J Sports Med*. 2016;50(5):273–280.
- [8] Gottschall JS, Kram R. Ground reaction forces during downhill and uphill running. *J Biomech*. 2005;38(3):445–452.
- [9] Hulme A, Nielsen RO, Timpka T, Verhagen E, Finch C. Risk and protective factors for middle- and long-distance running-related injury. *Sports Med*. 2017;47(5):869–886.
- [10] Impellizzeri FM, Tenan MS, Kempton T, Novak A, Coutts AJ. Acute:chronic workload ratio: conceptual issues and fundamental pitfalls. *Int J Sports Physiol Perform*. 2020;15(6):907–913.
- [11] Karvonen MJ, Kentala E, Mustala O. The effects of training on heart rate; a longitudinal study. *Ann Med Exp Biol Fenn*. 1957;35(3):307–315.
- [12] McHugh MP. Recent advances in the understanding of the repeated bout effect: the protective effect against muscle damage from a single bout of eccentric exercise. *Scand J Med Sci Sports*. 2003;13(2):88–97.
- [13] Minetti AE, Moia C, Roi GS, Susta D, Ferretti G. Energy cost of walking and running at extreme uphill and downhill slopes. *J Appl Physiol*. 2002;93(3):1039–1046. doi:10.1152/japplphysiol.01177.2001
- [14] Neal BS, Bramah C, McCarthy-Ryan MF, Moore IS, Napier C, Paquette MR, Gruber AH. Using wearable technology data to explain recreational running injury: a prospective longitudinal feasibility study. *Phys Ther Sport*. 2024;65:130–136. doi:10.1016/j.ptsp.2023.12.010
- [15] Nielsen RO, Parner ET, Nohr EA, Sørensen H, Lind M, Rasmussen S. Excessive progression in weekly running distance and risk of running-related injuries: an association which varies according to type of injury. *J Orthop Sports Phys Ther*. 2014;44(10):739–747. doi:10.2519/jospt.2014.5164
- [16] Paquette MR, Napier C, Willy RW, Stellingwerff T. Moving beyond weekly “distance”: optimizing quantification of training load in runners. *J Orthop Sports Phys Ther*. 2020;50(10):564–569. doi:10.2519/jospt.2020.9533
- [17] Plews DJ, Laursen PB, Stanley J, Kilding AE, Buchheit M. Training adaptation and heart rate variability in elite endurance athletes: opening the door to effective monitoring. *Sports Med*. 2013;43(9):773–781. doi:10.1007/s40279-013-0071-8
- [18] Seiler S. What is best practice for training intensity and duration distribution in endurance athletes? *Int J Sports Physiol Perform*. 2010;5(3):276–291.
- [19] Soligard T, Schweltnus M, Alonso JM, et al. How much is too much? (Part 1) International Olympic Committee consensus statement on load in sport and risk of injury. *Br J Sports Med*. 2016;50(17):1030–1041. doi:10.1136/bjsports-2016-096581
- [20] Strava. Relative Effort. Strava Help Center, support.strava.com/en-us/articles/15401794-relative-effort (accessed 25 Sep 2026).
- [21] Tanaka H, Monahan KD, Seals DR. Age-predicted maximal heart rate revisited. *J Am Coll Cardiol*. 2001;37(1):153–156. doi:10.1016/S0735-1097(00)01054-8
- [22] Vernillo G, Giandolini M, Edwards WB, et al. Biomechanics and physiology of uphill and downhill running. *Sports Med*. 2017;47(4):615–629. doi:10.1007/s40279-016-0605-y